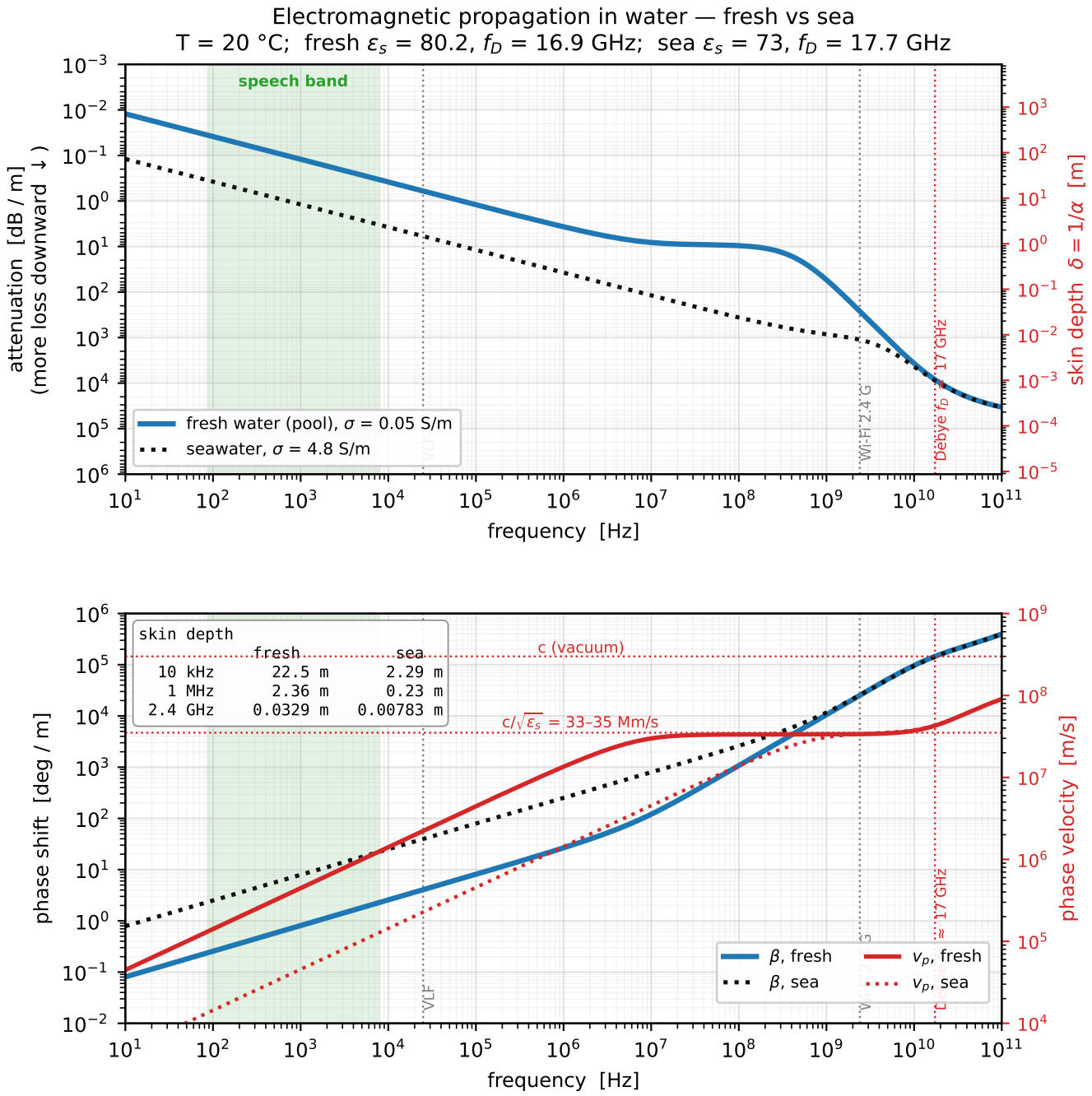

Electromagnetic propagation in water — fresh vs sea
T = 20 °C; fresh ε_s = 80.2, f_D = 16.9 GHz; sea ε_s = 73, f_D = 17.7 GHz
speech band
attenuation [dB / m]
(more loss downward ↓)
skin depth $\delta = 1/\alpha$ [m]
fresh water (pool), σ = 0.05 S/m
seawater, σ = 4.8 S/m
VLF
Wi-Fi 2.4 G
Debye f_D ≈ 17 GHz
frequency [Hz]
skin depth
fresh sea
10 kHz 22.5 m 2.29 m
1 MHz 2.36 m 0.23 m
2.4 GHz 0.0329 m 0.00783 m
c (vacuum)
$c/\sqrt{\varepsilon_s}$ = 33–35 Mm/s
phase shift [deg / m]
phase velocity [m/s]
β, fresh
β, sea
v_p, fresh
v_p, sea
VLF
≈ 17 GHz
frequency [Hz]